

General Purpose Ultrasound Phantom User Guide

Model 054GS



SUN NUCLEAR
A MIRION MEDICAL COMPANY

General Purpose Ultrasound Phantom User Guide

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Contents

Introduction	1	Inspection	15
Overview	1	Repair	15
Accessories	1	Cleaning	15
Key Tests with Model 054GS	2	Storage	15
Performing Measurements	3	Disposal and Recycling	16
Instructions for Using Removable		Contacting Sun Nuclear Support	16
Water Well and Covers	3	Support Website	16
General Guidelines for Performing		Specifications	17
Measurements	4	Product Specifications	17
Establishing a Baseline	4	Target Layout	17
Testing Procedures	5	Phantom	18
Uniformity Testing	5	Wire Targets	18
Depth of Penetration Testing	6	Near Field Group	18
Beam Profile, Focal Zone and		Vertical Distance Group	18
Lateral Response Width	7	Horizontal Distance Group	18
Vertical Distance Measurements	8	Axial/Lateral Distance Groups	19
Horizontal Distance		Anechoic Cylinders	19
Measurements	8	Gray Scale Targets	19
Axial and Lateral Resolution		Zerdine	21
Testing	9	QA Record for Model 054GS	23
Elevational Testing	11	Regulatory Supplement	25
Low-Contrast Target		Sun Nuclear Corporation Symbols	25
Detectability	12	Operator Responsibility	26
Grayscale Contrast Sensitivity	12	Reporting Health or Safety	
Dead Zone Assessment	13	Related Issues or Concerns	26
Support and Maintenance	15	Modifications to Equipment	26
Hardware Maintenance	15		

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1 Introduction

Overview

The Model 054GS General Purpose Ultrasound Phantom is a sturdy, reliable phantom for testing the imaging performance of ultrasonic systems.

The phantom is made of Sun Nuclear's proprietary Zerdine® hydrogel polymer, which has been formulated to provide tissue mimicking properties including compatibility with harmonic imaging. To maximize phantom lifetime, this gel is contained in a rugged ABS plastic housing with a Saran-based laminate membrane.

Sun Nuclear is certified to ISO 9001:2008 standards. We have an in-house test facility to measure acoustic

properties of materials. In addition, ultrasound imaging systems are used to inspect each phantom. Every ultrasound phantom Sun Nuclear distributes has passed thorough testing during manufacture and completion to ensure the highest quality product available. A Certificate of Compliance is issued with each phantom.

Additional guidance on establishing a quality assurance program can be found in the accreditation programs established by the ACR and AIUM at www.acr.org or www.aium.org.

Accessories

- Removable water well
- Removable endocavity cover
- Removable storage cover
- Carry case
- Certificate of Compliance
- Model 054GS User Guide
- QA worksheet

Key Tests with Model 054GS

For more information on these tests, see *Testing Procedures* on page 5.

- Uniformity
- Depth of Penetration
- Beam Profile/ Focal Zone/ Lateral Response Width
- Vertical Distance Measurement
- Horizontal Distance Measurement
- Axial and Lateral Resolution
- Elevational Resolution
- Low-Contrast Target Detectability
- Grayscale Contrast Sensitivity
- Dead Zone Assessment

2 Performing Measurements

Instructions for Using Removable Water Well and Covers

The phantom is shipped with the protective cover attached to the phantom. This can be removed by stretching the elastic latches on either side of the phantom up and off of the protective cover. The included water well and covers are easily secured to the phantom with these same rubber latches. Simply place the water well or cover on top of the phantom and stretch the elastic latches up and over the attachment point on either side of the accessory.



Cover on for storage



Attach cover with latches

Figure 2-1. Cover Secured with Latches

Coupling gel can be applied directly to the scan surface. This option is best used with linear transducers. For curved arrays, the water well may be attached and filled with water to provide better coupling. Side Fire transducers can be particularly challenging to scan with a standard phantom. Sun Nuclear has designed a removable endocavity cover for these transducers. When this accessory is attached, the phantom should be placed on its back and the cover should be filled with water.



Water well for coupling curved probes



Endocavity well

Figure 2-2. Water Well and Endocavity Well Accessories

When finished scanning, it is best to clean the scan surface of any water or coupling gel and replace the protective cover.

General Guidelines for Performing Measurements

It is recommended that all measurements be performed at the most frequently used imaging arrangements. The importance of these tests is to make sure that system performance remains constant over an extended period of time.

Measurements may also be used to compare the performance of various setups of the same machine or to compare different machines in a quantitative manner.

The following are general steps for imaging all targets:

- Some wires will appear as short lines rather than dots. When using the electronic calipers, always take measurements from a point on one echo to the same point on the next (i.e., center to center). Otherwise, errors may be introduced.
- If a convex probe is used, center the target within the scan plane in order to minimize degradation and distortion introduced on the outer edges of the probe.
- When assessing vertical distance measurements, DO NOT press on the scanning surface. Pressure on the scanning surface causes the wires to become temporarily displaced, making vertical distance measurements inaccurate.
- When assessing horizontal distance accuracy, ensure that the scan plane is perpendicular to the horizontal target group. Rotation of the probe will result in inaccurate distances.
- Always be sure the phantom is scanned while at room temperature. A phantom just received may be colder or hotter than room temperature depending on where it was stored during shipping. Temperature affects the speed of sound and, ultimately, the perceived measurements. The phantom should be stored at room temperature for at least 24 hours before use to ensure its core temperature is correct.
- The most accurate measurements will be made with the phantom $22^{\circ}\text{C} \pm 1^{\circ}\text{C}$ (70°F – 73°F).

Establishing a Baseline

Before performing routine quality assurance measurements, establish:

- 1 System settings for each measurement:
System setup can have a dramatic impact on the results obtained from quality assurance measurements. You must establish and record what system settings should be used for each of the quality assurance tests. These same settings should be used each time the test is performed. If not, then the conclusions drawn may not be valid. Sun Nuclear recommends that you use the most commonly used settings for the type of probe tested- i.e., the liver preset values for an abdominal probe- which are called a “normal” technique in the sections that follow.

2 Baseline measurements:

The first set of measurements taken will be the baseline measurements for the combination of system settings and phantom. Record the system settings and phantom serial number used to acquire each measurement along with your measurement results. On subsequent scans, refer to the baseline results to determine if the ultrasound system has drifted to an unacceptable level. It is each facility's responsibility to establish the magnitude of drift allowed before corrective action is warranted.

3 Allowable deviation from baseline measurements:

measurement should be calculated and recorded. At some point the difference will be large enough that some action is required (call service, replace system, etc.). Each facility needs to determine the action level for each test. You should refer to the user's manual of your ultrasound scanner and note the stated accuracies of the system's general imaging measurements. These stated accuracies may greatly influence the conclusion made when evaluating the ultrasound system. For example, if the measurement accuracy for your system is 10% for distances up to 2 cm, the scanner may detect 2.0 cm as being anywhere from 1.8 cm to 2.2 cm and still be functioning properly. The user is responsible for establishing action levels.

4 Frequency of system assessment:

How often each system is evaluated is also up to each facility to determine. Sun Nuclear recommends at least annually.

Reference the accreditation programs established by the ACR and AIUM at www.acr.org or www.aium.org for further guidance on establishing a QA program.

Testing Procedures

The following sections outline procedures for routine quality control tests with the Model 054GS. It may be useful to refer to the target map, shown in the Specifications section of page 14, when reviewing these procedures.

Uniformity Testing

Uniformity is defined as the ability of the machine to display echoes of the same magnitude and depth with equal brightness on the display. This is a good test to ensure all crystals within the transducer are functioning.

- 1** Apply coupling gel to the scanning surface or fill the water trough with tap water
- 2** Position the transducer on the scanning surface in a region with a minimum number of targets.
- 3** Adjust the instrument settings (gain, TGC, output, etc.) as for a "normal" technique. Record these settings for use on subsequent testing.
- 4** Align the probe so that the targets are maximized.
- 5** Freeze the image and obtain a hard copy.

- 6 Observe the general appearance of the phantom. Note if all regions at the same depth are displayed with the same intensity across the image.
- 7 Record your observations.

Depth of Penetration Testing

Depth of penetration, also called maximum depth of visualization or sensitivity, is the greatest distance in a phantom for which echo signals caused by scattering in the background material can be detected on the display. The depth of penetration is determined by the frequency of the transducer, the attenuation of the medium being imaged and the system settings.



Note: To register accurate vertical distance measurements, **DO NOT APPLY PRESSURE TO THE SCANNING SURFACE!** Sun Nuclear strongly encourages the user to scan the phantom with the water well filled with water or coupling gel so the transducer does not make direct contact with the scanning surface. As with a patient, even the slightest amount of pressure on the scanning surface will cause incorrect distances to be measured.

- 1 Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2 Position the transducer to acquire an image of a vertical plane target. (The wires should appear as dots, not lines).
- 3 Adjust the instrument settings (gain, TGC, output, etc.) as for a “normal” technique. Record these settings for use on subsequent testing.
- 4 Align the probe so that all the vertical targets are displayed at their maximum intensity level.
- 5 While actively scanning, look to see where the backscattered echoes within the background material disappear. Be careful not to confuse electronic noise with the background backscattered echoes. Electronic noise will move but back-scattered echoes will remain stationary while maintaining the transducer in a fixed position.
- 6 Freeze the image.
- 7 With electronic calipers measure the distance between the scanning surface and the last identifiable echoes due to scattering.



Note: Usually the wires stay visible even though the back scattered echoes are not. Remember to measure the distance to the scattered echoes, not to the last visible wire.

- 8 Record this distance on a record sheet and compare with baseline depth.

Beam Profile, Focal Zone and Lateral Response Width

The beam profile is the shape of the ultrasound beam. A typical beam profile is shown in Figure 2-3. The narrowest region within the beam profile is indicative of the focal point. By convention, the region surrounding the focal point with intensity within 3 dB of maximum is the focal zone. The best images are obtained while within the focal zone. The vertical wire target group is useful for determining the beam profile and the focal zone of a system, as follows:

- 1 Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2 Position the transducer in a vertical plane. (The wires should appear as dots, not lines).
- 3 Adjust the instrument settings (gain, TGC, output, etc.) as for a “normal” liver technique. Record these settings for use on subsequent testing.
- 4 Align the probe so that all the vertical targets are displayed at their maximum intensity level to ensure the transducer is imaging a vertical plane.
- 5 Freeze the image and obtain a hard copy.



Note: Some of the targets will appear as short horizontal lines rather than dots on the frozen image.

- 6 Measure the horizontal length of the targets. These measurements represent the lateral response width of the system at the different depths and setup. The minimum length is indicative of the location of the focal point.

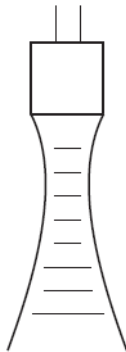


Figure 2-3. Typical Beam Profile

- 7 If a smooth curve is drawn to connect the edges of the targets, the beam profile is easily discernible.
- 8 If using a variable focused transducer, repeat the above procedure for several different focal zones (those settings most commonly used are recommended).

- 9 Record the focal point and save the hard copy.

Vertical Distance Measurements

A vertical distance is defined as the distance along the axis of the beam. The vertical wire targets are used to assess the accuracy of vertical distance measurements as follows:

- 1 Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2 Position the transducer in a vertical plane. (The wires should appear as dots, not lines). Do not apply excessive pressure as this may temporarily compress the target and skew the measurements.
- 3 Adjust the instrument settings (gain, TGC, output, etc.) as for a “normal” technique. Record these settings for use on subsequent testing.
- 4 Align the probe so that all the vertical targets are displayed at their maximum intensity level.
- 5 Freeze the image and obtain a hard copy.
- 6 Using electronic calipers, measure the distances between two wires at various depths or align the echoes to the display markers for comparison.
- 7 Record these measurements.
- 8 Compare the measured values with the recorded baseline distances.

Horizontal Distance Measurements

The horizontal target group is used to determine the accuracy of measurements made perpendicular to the beam axis. There are two horizontal plane target groups. The 4 cm deep group contains 4 wires with spacing of 10 mm and 20 mm. The 9 cm deep group has 20 mm spacing between each of the 7 wires. Refer to the target diagram attached to your phantom. Testing is performed as follows:

- 1 Fill the water trough with tap water.
- 2 Position the transducer in a vertical plane. (The wires should appear as dots, not lines).
- 3 Adjust the instrument settings (gain, TGC, output, etc.) as for a “normal” technique. Record these settings for use on subsequent testing.
- 4 Align the probe so that all the horizontal targets are displayed at their maximum intensity level.
- 5 Freeze the image and obtain a hard copy.
- 6 Using electronic calipers, measure the distances between two wires along the horizontal plane.
- 7 Record these measurements.

- 8 Compare the measured values with the known distances between the targets.

Axial and Lateral Resolution Testing

Axial resolution is defined as the ability of an ultrasound system to resolve objects in close proximity along the axis of the beam. In other words, it determines how close two objects can be along the axis of the beam and still be detected as two distinct objects. Axial resolution is proportional to the length of the system's transmitted ultrasonic pulse or pulse length.

Lateral resolution is similar to axial resolution except it is concerned with the resolution perpendicular to the beam axis. Lateral resolution will improve with a narrowing of the beam width. Therefore, within the focal zone, the lateral resolution will be at its best.

The Model 054GS has two combined axial and lateral resolution target groups. The first group, at a depth of 3 cm, is designed for probes of 5 MHz and above. It consists of 13 parallel nylon wires of 80 microns diameter. The second target is located at 11 cm depth for evaluation of low frequency probes. It consists of 11 parallel nylon wires of 80 microns diameter.

To measure axial and lateral resolution, refer to Figures 2 and 3 for the layout of the axial/lateral target groups and perform the following steps:

- 1 Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2 Position the transducer above the axial resolution targets in a vertical plane. (The wires should appear as dots, not lines).
- 3 Adjust the instrument settings (gain, TGC, output, etc.) as for a "normal" technique. Record these settings for use on subsequent testing.
- 4 Align the probe so that all the targets are displayed at their maximum intensity level.
- 5 Freeze the image and obtain a hard copy.
- 6 Examine the image to determine the last pair of wires to be distinguished as two separate entities. If the last pair of wires to be resolved is separated by a distance of 1 mm then record the axial resolution as being "in between 0.5 mm and 1.0 mm".

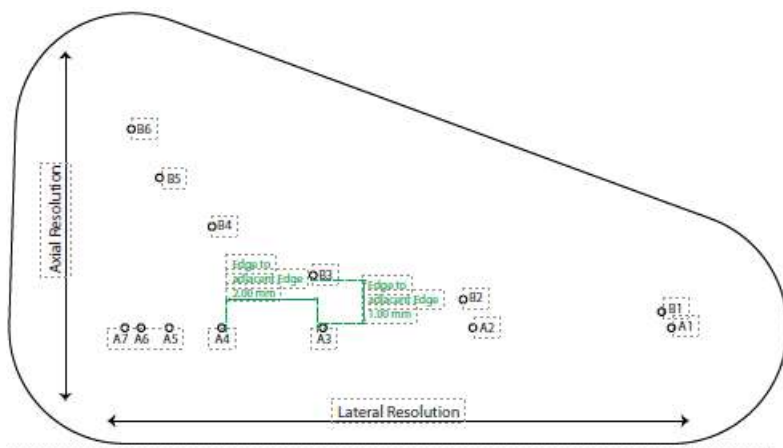


Figure 2-4. Combined Axial/Lateral Resolution Targets at 3 cm Depth (Top)

Table 2-1. Table Listing Distances Between Axial/Lateral Resolution Targets

	Targets					
	A1-B1	A2-B2	A3-B3	A4-B4	A5-B5	A6-B6
Axial Resolution (mm)	.25	.5	1.0	2.0	3.0	4.0
	Targets					
	A1-A2	A2-A3	A3-A4	A4-A5	A5-A6	A6-A7
Lateral Resolution (mm)	4.00	3.0	2.0	1.0	0.5	0.25

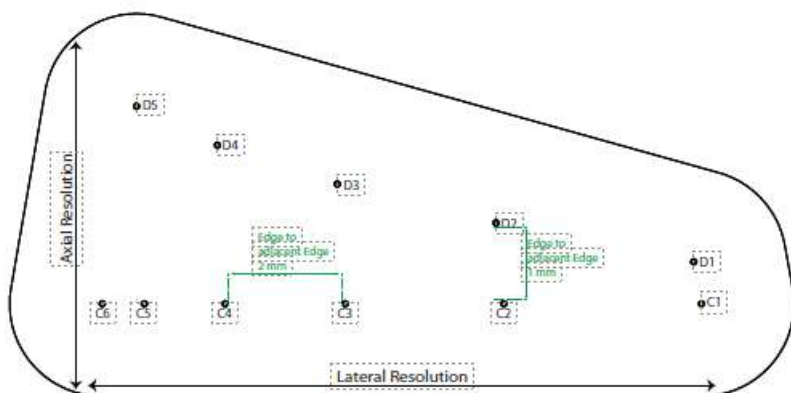


Figure 2-5. Combined Axial/Lateral Resolution Targets at 11 cm Depth

Table 2-2. Table Listing Distances Between Axial/Lateral Resolution Targets

	Targets				
	C1-D1	C2-D2	C3-D3	C4-D4	C5-D5
Axial Resolution (mm)	1.0	2.0	3.0	4.0	5.0
	Targets				
	C1-C2	C2-C3	C3-C4	C4-C5	C5-C6
Lateral Resolution (mm)	5.0	4.0	3.0	2.0	1.0

Elevational Testing

A full characterization of system resolution requires measurement of elevational resolution, or slice thickness. Slice thickness is typically much coarser than axial and lateral resolution as most ultrasound transducer arrays are mechanically focused in the thickness dimensions.

Elevational resolution may be estimated using a method first described by Skolnick¹. It uses the vertical wire targets as follows:

- 1 Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2 Adjust the instrument settings (gain, TGC, output, etc.) as for a “normal” technique. Record these settings for use on subsequent testing.
- 3 Orient the transducer to image the length of the vertical target wires, taking care to adjust the tilt so that the wires are lined up in a vertical column.

1. Skolnick, ML. “Estimation of ultrasound beam width in the elevation (section thickness) plane.” *Radiology*. 1991 Jul;180(1):286-8

- 4** Rotate the transducer 45° so that only a partial length of the wires is now visible.
- 5** Freeze the image and measure the length of each wire segment with the electronic calipers.
- 6** Record the measurements.

Low-Contrast Target Detectability

Machines have a tendency to represent low-contrast structures smaller than they actually are and with irregular rather than smooth borders. This is referred to as fill-in. Ideally, the fill-in effect will be minimal.

In the Model 054GS, five cylinders having no scatter are provided in the phantom to test a machine's ability to image cyst-like structures over range of depths. The cylinders are 8 mm in diameter and located at depths of 4, 7, 10, 13, and 16 cm. Refer to the target diagram attached to your phantom. The accuracy of the machine's representation of the mass (proper size and shape) may be determined using this target. Because of the low attenuation in this mass, you may notice enhancement behind the target.

- 1** Testing for low-contrast target detectability is performed as follows:
- 2** Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 3** Position the transducer above the cyst of interest and perpendicular to the wires. You should be imaging the circular cross section of the cylinders.
- 4** Adjust the instrument settings (gain, TGC, output, etc.) as for a "normal" technique. Record these settings for use on subsequent testing.
- 5** Align the probe so that the target is maximized.
- 6** Freeze the image and obtain a hard copy.
- 7** Observe the general appearance of the tumor. Note if you are able to see the mass
- 8** A more detailed analysis can be performed by measuring the width and height of each mass.
- 9** Record your observations.

Grayscale Contrast Sensitivity

In the Model 054GS, six cylinders having contrast ranging from anechoic to hyperechoic with respect to the background material are provided in the phantom to test a machine's dynamic range. The cylinder diameter is 8 mm at a depth of 4 cm. These masses are useful in determining the ultrasonic system's capability of distinguishing targets of varying gray scale levels. The accuracy of the machine's representation of the mass (proper size and shape) may also be determined using this target. Refer to target diagram attached to your phantom.

The dynamic range of an ultrasound imager can be evaluated using the gray scale masses in conjunction with the cystic masses and the hyperechoic masses. Testing of grayscale contrast is performed as follows:

- 1** Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2** Position the transducer above the tumor of interest and perpendicular to the wires. (The tumor should appear as a circular region).
- 3** Adjust the instrument settings (gain, TGC, output, etc.) as for a "normal" technique. Record these settings for use on subsequent testing.
- 4** Align the probe so that the target is maximized.
- 5** Freeze the image and obtain a hard copy.
- 6** Observe the general appearance of each tumor. Note if you are able to see each of the masses.
- 7** A more detailed analysis can be performed by measuring the width and height of each mass.
- 8** Record your observations.

Dead Zone Assessment

The near field group is used to assess the distance from the front face of the transducer to the closest identifiable echo. This region, where no useful information is obtained, is commonly referred to as the "dead-zone," "ring-down distance," or "near field resolution." The dead zone occurs because the ultrasound system can-not send and receive data simultaneously. It is instrument dependent and is diminished as frequency is increased. A change in your system's dead zone is indicative of a problem with the transducer, the pulsing system or both.

The near field group consists of parallel, 100 micron diameter, nylon, monofilament wires horizontally spaced 6 mm apart from center to center (Figure 2-6). Vertical distance from the center of each wire to the top edge of the scanning surface ranges from 5 mm down to 1 mm in 1 mm increments.

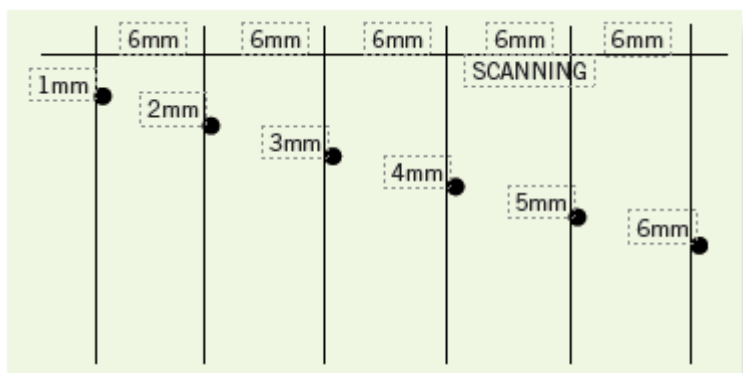


Figure 2-6. Near Field Target Spacing

The depth of the dead zone may be measured as follows:

- 1** Apply coupling gel to the scanning surface or fill the water trough with tap water.
- 2** Position the transducer above the near field resolution target and perpendicular to the wires. (The wires should appear as dots, not lines).
- 3** Adjust the instrument settings (gain, TGC, output, etc.) to maximize resolution in the near field. Record these settings for use on subsequent testing.
- 4** Freeze the image while the near field targets are clearly displayed.
- 5** Count how many wires of the near field target you can see. Subtracting this number from the total number of targets gives you the dead zone measurement. For instance, if 3 targets are visible, the dead zone distance = 3 mm (6mm-3mm).

An alternative method uses the electronic calipers to measure the distance between the transducer face and the closest wire target to be resolved from the reverberation. If the first target to be resolved is at 4 mm, then the dead zone distance is "something less than 4 mm".

- 6** Record this distance and compare with baseline measurements.

3 Support and Maintenance

Hardware Maintenance

With proper care, the Model 054GS will withstand years of normal use. Below are some guidelines to follow.

Inspection

Inspect your phantom regularly for signs of damage and weight loss. If any noticeable changes to the phantom are detected, return the phantom IMMEDIATELY for repair or replacement.



Note: *At least once a year, weigh your phantom and compare to original weight noted on certificate of compliance. If the phantom has lost or gained more than 1% of its original weight, or if the scan surface appears depressed, contact Sun Nuclear Support.*

Repair

The scanning surface is the most important item on the phantom to protect. It can withstand normal scanning pressure but DO NOT press on the scanning surface with your fingernails or any other sharp objects. If the scanning surface becomes damaged, seal the phantom in an airtight container and IMMEDIATELY contact Sun Nuclear Support for return authorization.

Cleaning

The phantom may be cleaned with mild soap and water ONLY. Avoid solvent-based, alcohol-based, or abrasive cleaning agents.

Storage

For longest life, the phantom should be cleaned after each use and stored at room temperature in the provided carry case. The primary concern is gel desiccation due to loss of water vapor through the membrane. In addition, the thermal stresses associated with the freeze/thaw cycle may cause the gel to crack or damage the housing integrity, while extreme heat may accelerate water vapor transmission through the membrane. To minimize desiccation, always store the phantom in the air-tight carry case with the removable storage cover attached.



CAUTION: This product contains Zerdine, a non-flowing, water-based, polyacrylamide material which is fully sealed within the phantom housing. Zerdine contains trace amounts of the residual monomer acrylamide CAS#79-06-1. There are no known hazards when the phantom is used and stored as intended. Zerdine is fully cured and will not leak from the housing. Damage to the integrity of the housing may expose the user to trace amounts of acrylamide monomer. The amount is not sufficient to pose an acute health risk, but it is still advised to wear protective gloves if handling exposed Zerdine gel due to the potential long-term hazards of the monomer. It is also advisable to wash hands and all surfaces with soap and water after handling exposed Zerdine gel.

Disposal and Recycling



Regulations regarding disposal of materials with trace acrylamide monomer vary by locality. Contact your local authority for instructions. If assistance is desired in the proper disposal of this product, including accessories and components, after its useful life, please return to Sun Nuclear.

Contacting Sun Nuclear Support

You may request support in two ways:

- Submit a support request using our online form. See *Support Website* below.
- Contact the Sun Nuclear Support team by telephone:
 - U.S.A.: +1 321-259-6862, Option 3
 - Netherlands: +31 20 399 90 41, Option 1
 - Germany: +49 61 02 50 49 500, Option 2

Support Website

- 1 Open an internet browser and navigate to sunnuclear.com/support.
- 2 Enter your email address and password and then click **Login**.
 - To download product information, click **Products and Devices**, select the product, and then select the download type.
 - To open a Support request, click **Open New Case**, complete the form, and then click **Create Case**.

4 Specifications

Product Specifications



Note: All dimensions without tolerances are nominal

Note: All measurements made at $22^{\circ} \pm 1^{\circ} \text{ C}$

Target Layout

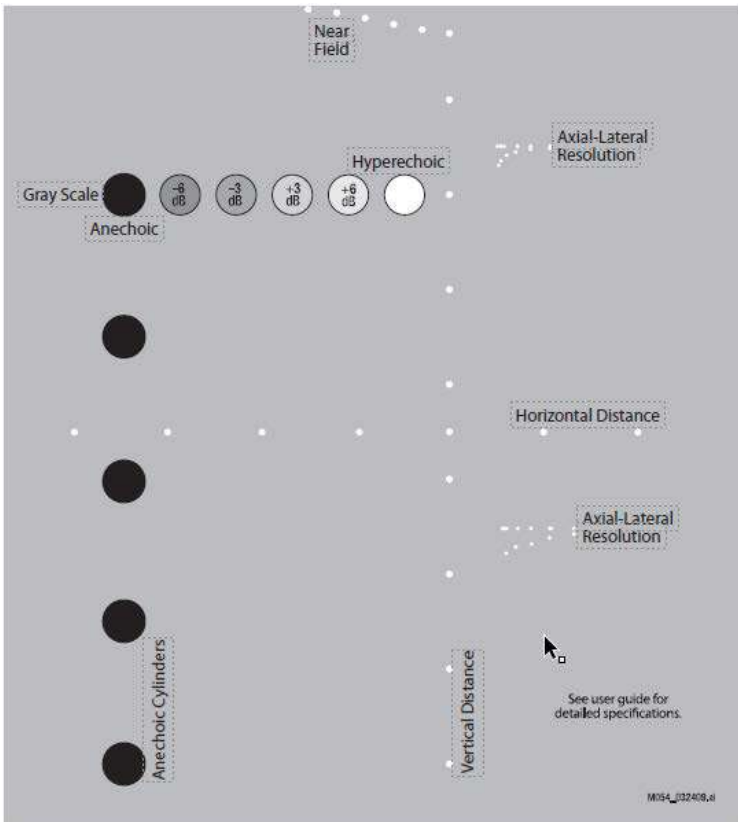


Figure 4-1. Target Layout

Phantom

Table 4-1. Phantom specifications

Characteristic	Specification
Housing	ABS Plastic
Outer Dimensions	17.8 x 12.7 x 20.3 cm (7 x 5 x 8")
Scanning Surface	14 x 9 cm Saran-based laminate
Scanning Material	Zerdine® tissue mimicking gel
Speed of Sound	1540 m/s
Other	Compatible with harmonic imaging

Wire Targets

Table 4-2. Wire Targets Specifications

Characteristic	Specification
Material	Nylon Monofilament

Near Field Group

Table 4-3. Near Field Group Specifications

Characteristic	Specification
Number of Targets	6
Diameter	100 microns
Depth range	2 to 16 mm
Vertical distance between targets	20 mm

Vertical Distance Group

Table 4-4. Vertical Distance Group Specifications

Characteristic	Specification
Number targets	8
Diameter	100 microns
Depth range	1 to 6 cm
Vertical distance between targets	1 mm

Horizontal Distance Group

Table 4-5. Horizontal Distance Group Specifications

Characteristic	Specification
Number of groups	1
Diameter	100 microns
Depths	9 cm

Table 4-5. Horizontal Distance Group Specifications

Characteristic	Specification
Number of targets	7 respectively
Horizontal distance between targets	20 mm respectively

Axial/Lateral Distance Groups

Table 4-6. Axial/Lateral Distance Group Specifications

Characteristic	Specification
Group 1	
Diameter	80 microns
Depths	3 cm
Axial & Lateral separation between targets	4, 3, 2, 1, 0.5, & 0.25 mm
Group 2	
Diameter	80 microns
Depths	11 cm
Axial & Lateral separation between targets	5, 4, 3, 2, & 1 mm

Anechoic Cylinders

Table 4-7. Anechoic Cylinders Specifications

Characteristic	Specification
Number of Cylinders	5
Contrast	Anechoic
Diameter	8 mm
Depths	4, 7, 10, 13 & 16 cm

Gray Scale Targets

Table 4-8. Gray Scale Targets

Characteristic	Specification
Number of Targets	6
Contrasts	Anechoic, -6dB, -3 dB, +3 dB, +6dB, & Hyperechoic
Diameter	8 mm
Depth	4 cm

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5 Zerdine

Each phantom in the Model 054GS set is constructed from a patented, solid elastic material developed at Sun Nuclear called Zerdine. Phantoms constructed from Zerdine will not melt or leak when punctured, and they do not require refrigeration. Zerdine is also more elastic than other materials and allows more pressure to be applied to the scanning surface without subsequent damage to the material. At normal room temperatures, Zerdine will accurately simulate the ultrasound characteristics found in human liver tissue. Specific proprietary fabrication procedures enable close control over the homogeneity of Zerdine and the reliability of its acoustic characteristics from batch to batch.

The formulation system established at Sun Nuclear is geared to independently control:

- The speed of sound in the optimal range of 1510 to 1700 m/s.
- Attenuation in the optimal range of 0.05 and 1.5 dB/cm/MHz.
- Scatter or relative contrast in the optimal range of -15 to +15 dB in relation to a scatter baseline equivalent to human liver tissue.
- Elasticity with Young's Modulus in the optimal range of 4 to 90 kPa.

At normal room temperature, the Zerdine response to ultrasonic excitations will simulate the ultrasonic response of human tissue. The relation between the acoustic attenuation, A , and the acoustic frequency, F , is of the form $A = A_0 F^n$ with values of the power coefficient, n , in the range of 0.8 to 1.10, indicating the proportional increase of the acoustic attenuation with frequency. Backscatter characteristics can be adjusted through the addition of predetermined amounts of calibrated scatter material, and are fully compatible with harmonic imaging. Zerdine can be molded into very intricate shapes, and the material can be cured in layers allowing the production of "multi-tissue" phantoms. Zerdine, like most other phantom materials, will desiccate if unprotected; thus, all phantoms must be stored properly. If stored in the case provided, your phantom should last many years.

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Appendix A: QA Record for Model 054GS

GENERAL PURPOSE ULTRASOUND PHANTOM QUALITY ASSURANCE RECORD

Location: _____ Unit: _____ Probe: _____ QC Phantom SN: _____

Machine Settings:

Depth of Field (FOV) _____ cm Gain: _____ Power: _____

Focal Zone(s) _____ cm _____ cm _____ cm _____ cm _____ cm

Preprocessing _____ Post Processing _____ Dynamic Range _____

Other: _____

TEST		BASELINE REMARKS	TEST RESULTS	VARIANCE	COMMENTS
Uniformity					
Depth of Penetration					
Beam Profile/ Focal Zone/ Lateral Width Response					
Vertical Distance					
Horizontal Distance					
Axial and Lateral Resolution	3 cm				
	11 cm				
Elevational Resolution					
Low-Contrast Target Detectability					
Gray Scale Contrast Sensitivity					
Near Field					

Duplicate as Needed
One Sheet Per System Setup

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Appendix B: Regulatory Supplement

In addition to the regulatory information contained in the body of this manual, the following supplemental regulatory information is provided.

Sun Nuclear Corporation Symbols

The following symbols are used in this guide and in Sun Nuclear Corporation's product labels.



WARNING: This symbol indicates a hazard that could result in major injury or equipment damage. (EN ISO 7010, W001)



CAUTION: This symbol indicates a potential hazard that could result in minor injury or equipment damage. (EN ISO 15223-1, 5.4.4)



CAUTION: This symbol indicates a pinch hazard. (EN ISO 7010, W024)



Note: *Important or supporting information.*



Manufacturer's Identification (name and address). (EN ISO 15223-1, 5.1.1)



Date of Manufacture. (EN ISO 15223-1, 5.1.3)



Temperature limitation. (EN ISO 15223-1, 5.3.7)



Humidity limitation. (EN ISO 15223-1, 5.3.8)



Atmospheric pressure limitation. (EN ISO 15223-1, 5.3.9)



Serial Number. (EN ISO 15223-1, 5.1.7)



Catalog Number. (EN ISO 15223-1, 5.1.6)



Consult instructions for use. This equipment must be used in accordance with the instructions in this manual. Read all instructions and safety labels before use. (EN ISO 15223-1, 5.4.3)



Do not throw in trash; dispose of in an environmentally friendly way. (EN 50419)

Operator Responsibility

The instructions in this manual are intended for trained clinical personnel. The operator is solely responsible for the accurate setup and use of the phantom.

Reporting Health or Safety Related Issues or Concerns

A notice to the user and/or patient that any serious incident that has occurred in relation to the device should be reported to the manufacturer and the competent authority of the Member State in which the user and/or patient is established.

To report any safety or health related issues or concerns regarding the use of Sun Nuclear products, contact Sun Nuclear directly.

Modifications to Equipment

Any changes or modifications to the device that are not expressly approved by Sun Nuclear Corporation could void your warranty.

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